Teardown Benchmark Study

Tipper body and hydraulics - floor wear, body structure and tipping gear
Subject tipper platform against two construction-duty benchmarks

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Bodies examined	3, stripped to bare structure
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Summary

Three tipper bodies were stripped and compared for floor wear resistance, structural mass and tipping gear. Floor wear in mining overburden duty is the dominant warranty exposure on this platform. A hardened-steel bolt-on floor liner, adding 42.0 kg, is recommended for the mining and quarry duty variants and is not recommended for construction or coal duty, where the payload penalty outweighs the wear benefit.

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1 Scope and duty cycles

This study covers the tipper body floor, side and front panels, the sub-frame, the tipping cylinder and its mountings, and the body hinge arrangement. The chassis below the sub-frame is excluded and is covered in CVOEM/VAVE/TD-BM/2024/015.

1.1 Duty cycles

Tipper wear is driven almost entirely by material type, not by load or by distance. Four duty cycles are distinguished in this study because the appropriate specification differs materially between them.

Duty	Material carried	Typical load (t)	Wear mechanism
Construction	Sand, aggregate, soil	18 to 22	Abrasive sliding during discharge
Quarry	Blasted rock, graded stone	20 to 25	Impact during loading, then abrasion
Mining overburden	Mixed rock and clay	22 to 26	Severe impact and abrasion, sticky material
Coal	Bulk coal	16 to 20	Low abrasion, corrosive moisture

Scope note. A single floor specification across all four duty cycles is either over-specified for coal and construction or under-specified for mining. The platform currently uses one specification.

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1.2 Measured wear

Floor thickness was measured on returned bodies at 100,000 km across the four duty cycles, at nine points on a grid, and the mean loss recorded. Measurements were taken on the subject platform's standard 3 mm floor and on benchmark bodies fitting a bolt-on liner.

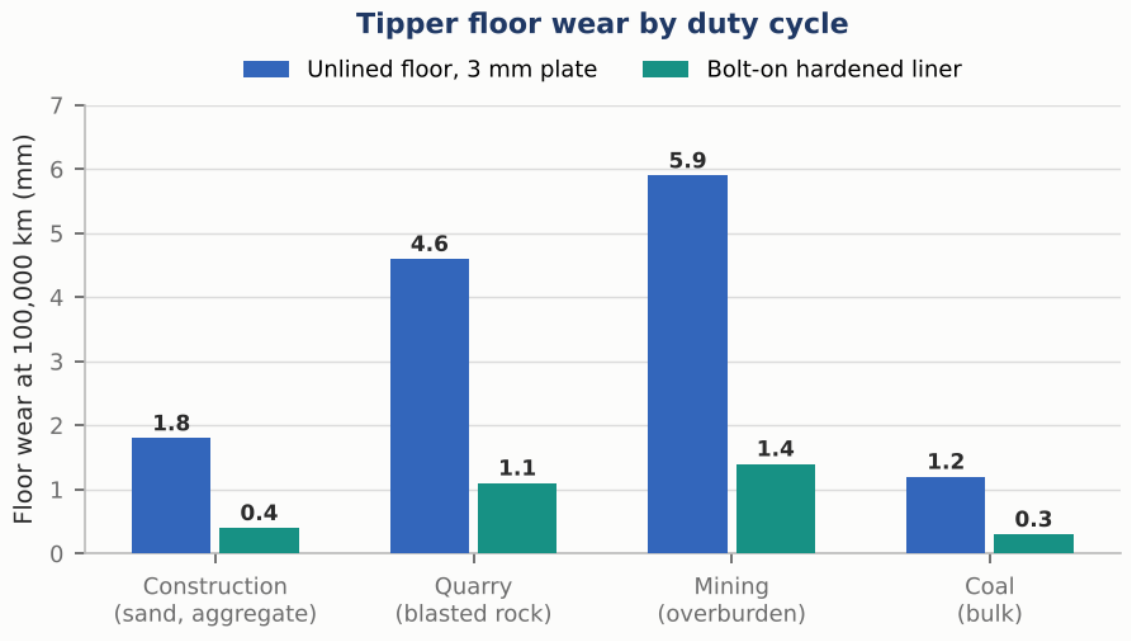


Figure 1 - Mean floor wear at 100,000 km by duty cycle, unlined against bolt-on hardened liner. Mining overburden is 3.3 times the construction rate.

Duty	Unlined wear (mm)	Lined wear (mm)	Floor life, unlined
Construction	1.8	0.4	Beyond vehicle life
Quarry	4.6	1.1	About 65,000 km
Mining overburden	5.9	1.4	About 51,000 km
Coal	1.2	0.3	Beyond vehicle life

Finding 1.1. The 3 mm unlined floor reaches its wear limit at about 51,000 km in mining overburden duty, well inside the warranty period. In construction and coal duty it outlasts the vehicle.

2 Tipper body and hydraulics

2.1 Floor protection - principal finding

Both benchmarks fit a bolt-on hardened-steel floor liner on their mining and quarry duty variants. A hardened-steel bolt-on floor liner, adding 42.0 kg, is recommended for the mining and quarry-duty variant to reduce floor-wear warranty exposure. It is not recommended for construction or coal duty.

Attribute	Subject (unlined)	Benchmark D	Benchmark E
Floor plate	3 mm mild steel	3 mm + 6 mm liner	4 mm + 5 mm liner
Liner material	-	Hardened, 400 HB	Hardened, 450 HB
Liner attachment	-	Bolt-on, 24 fixings	Bolt-on, 18 fixings
Added mass (kg)	0	42.0	38.4
Liner replaceable	-	Yes, in service	Yes, in service
Est. liner cost (INR)	-	28,400	31,200

Finding 2.1. A hardened-steel bolt-on floor liner, plus 42.0 kg, is recommended for the mining and quarry-duty variant to reduce floor-wear warranty exposure. The liner is replaceable in service, so the body is not scrapped when the wear surface is consumed.

2.2 Why bolt-on rather than welded

A welded wear plate is cheaper to fit and cannot be replaced without cutting. A bolt-on liner costs more initially and converts a body replacement into a liner replacement. On mining duty, where the wear surface is consumed within the vehicle's life, the bolt-on arrangement is the lower whole-life cost.

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2.3 Payload consequence

The 42.0 kg liner is a direct payload reduction. On a 24 tonne mining load that is 0.18 per cent, which is within the normal loading tolerance and is not a commercial objection. On construction duty, where the unlined floor already outlasts the vehicle, the same 42.0 kg is a payload loss for no benefit.

Duty	Liner recommended	Rationale
Construction	No	Unlined floor outlasts the vehicle; liner is payload loss
Quarry	Yes	Unlined life about 65,000 km, inside warranty
Mining overburden	Yes	Unlined life about 51,000 km; dominant warranty exposure
Coal	No	Low abrasion; corrosion protection is the relevant measure

Finding 2.2. The liner is a duty-specific specification, not a platform-wide change. A proposal to fit it across all variants should be rejected, and a proposal to delete it from mining duty should be rejected equally. Both have been received.

2.4 Hydraulics

All three vehicles use a front-mounted telescopic tipping cylinder of comparable rating. Measured masses are within 6.2 kg. The benchmark cylinders use a composite wear ring at the first stage; the subject platform uses a bronze ring. No cost or mass opportunity arises.

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3 Body structure

3.1 Side and front panels

The subject body uses 4 mm side panels with six external stiffeners per side. Both benchmarks use 3 mm panels with eight stiffeners. The benchmark approach is 31.4 kg lighter per body at comparable panel stiffness, because stiffener pitch is a more efficient route to panel stiffness than plate thickness.

Element	Subject (kg)	Benchmark D (kg)	Benchmark E (kg)
Floor plate	348.2	342.6	401.8
Side panels, pair	286.4	255.0	248.2
Front panel	92.6	84.1	88.4
Tailgate	118.4	112.2	109.6
Sub-frame	412.8	386.4	394.2
Body total	1,258.4	1,180.3	1,242.2

Finding 3.1. Thinner side panels with a closer stiffener pitch save 31.4 kg per body at comparable stiffness. It is a body tooling change and is classified as a platform change.

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4 Tipping gear

4.1 Hinge arrangement

The subject body uses a pinned hinge with a greased bronze bush. Benchmark D uses a maintenance-free composite bush. Field data shows hinge bush replacement at about 140,000 km on the subject platform; the benchmark publishes a no-maintenance claim to 400,000 km.

4.2 Sub-frame attachment

The subject sub-frame is bolted to the chassis at eight points through fabricated brackets. Both benchmarks use six points with a longer bearing surface. Fewer, larger attachments reduce local chassis stress concentration and remove two brackets per side.

Finding 4.1. A six-point attachment with a longer bearing surface removes four brackets and 11.2 kg, and reduces local frame stress. It requires a frame stress re-analysis and is a platform change.

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5 Opportunities and conclusions

Ref	Opportunity	Mass (kg)	Class	Applies to
CB-13-01	Bolt-on hardened floor liner	+42.0	Running change	Mining, quarry
CB-13-02	3 mm sides with closer stiffener pitch	-31.4	Platform change	All
CB-13-03	Six-point sub-frame attachment	-11.2	Platform change	All
CB-13-04	Composite hinge bush	-0.4	Running change	All

- Floor wear in mining overburden duty is the dominant warranty exposure, with the unlined floor reaching its limit at about 51,000 km.
- The bolt-on liner adds 42.0 kg and converts a body replacement into a liner replacement. It is duty-specific and should not be applied platform-wide.
- Body structural mass is best addressed through stiffener pitch rather than plate thickness, worth 31.4 kg.
- Finding 2.2 should be recorded in the idea register, since proposals have been received in both directions - universal fitment and universal deletion.

Recommendation. Release CB-13-01 for mining and quarry variants only, and CB-13-04 for all variants. Record the duty-specific scope explicitly so the liner is not later assessed as a platform-wide mass penalty.

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